ANTI-VIRAL DRUGS

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OVERVIEW

- Viruses are obligate intracellular parasites that use the host cell's metabolic pathways for reproduction and hence making Selective toxicity difficult to achieve

- This limits the number of potential sites for antiviral drug action

- It should clearly be stated that both antibacterial and antifungal drugs have little or no effect on viruses

- Effective antiviral compounds have been developed for the treatment of some viral infections like those against herpes virus infections, human immune deficiency virus (HIV), influenza and hepatitis

Major site of anti-viral drugs

- Different sites of action of antiviral drugs have been identified even as other targets are still currently being investigated

- Generally, most of the current antiviral drugs work against the following;

I. Acting on the endogenous nucleosides as anti metabolites thus leading to inhibition of viral replication
II. Entry inhibition
III. Un-coating inhibition
IV. Inhibition of release and spread of the viruses

- Critically speaking and in summary, antiviral drugs target viral enzymes, virus directed enzymes and virus specific steps

Sites of antiviral drugs

- Look out for pictoral presentation of anti-viral activity

Cases when antivirals are needed

- A number of antiviral infections are self limiting especially in immune competent patients

- However there are instances when antivirals may need to be prescribed such as;

I. **Immunocompromised individuals** - This may include patients presenting with viral infection while also having conditions or being in situations like HIV, cancer, patients receiving immune modulating drugs etc

II. **Infection with high morbidity or mortality** - these conditions include herpes zoster virus infections,

Classification of antiviral drugs

Anti – herpes drugs

- Acyclovir
- Valacyclovir
- Ganciclovir
- Valganciclovir
- Famciclovir
- Cidofovir
- Foscarnet
- Idoxuridine
- Trifluridine

Anti-influenza drugs

- Amantadine
- Rimantadine
- Oseltamivir
- Zanamivir

Anti-Hepatitis drugs/Non-selective antiviral drugs

Primarily for Hepatitis B

- Lamivudine
- Adefovir dipivoxil
- Tenofovir

Primarily for Hepatitis C

- Ribavirin
- Interferon alpha

Antiretroviral drugs (ARVs)

CLASS	DRUGS
Nucleoside reverse transcriptase inhibitors (NRTIs)	Zidovudine, Lamivudine, Didanosine, Stavudine, Abacavir, Emtricitabine, Tenofovir
Non-nucleoside RTI (NNRTIs)	Nevirapine, Efavirenz, Delavirdine
Protease Inhibitors (Common Suffix - navir)	Ritonavir, Lopinavir, Azatanavir, Indinavir, Nelfinavir, Saquinavir, Amprenavir
Entry (fusion) inhibitors	Enfuvirtide
CCR5 receptor inhibitor	Maraviroc
Integrase inhibitors	Raltegravir

Drugs for herpesvirus infections

HERPESVIRUS INFECTIONS

- All herpesviruses are DNA viruses e.g. herpes simplex virus (HSV), varicella zoster virus (VZV) and cytomegalovirus (CMV)
- HSV often causes herpes genitalias (genital herpes) and herpes labialis (infection of the lips and mouth), herpetic keratoconjuctivitis (infection of the cornea and conjunctiva) and less commonly herpetic encephalitis which is a potentially fatal disease
- VZV causes chicken pox (Varicella) and shingles (herpes zoster) with chicken pox mostly occurring in children while shingles mostly occur in adults resulting from the activation of latent VZV in the dorsal root
- In patients with shingles, pain and lesions occur where the virus travels peripherally along sensory nerves to the corresponding cutaneous or mucosal surfaces
- The skin lesions eventually heal but can leave residual scars while post herpetic neuralgia is a common and disabling complications of shingles

Cytomegalovirus

- CMV in immunocompetent individuals are usually asymptomatic

- Symptomatic CMV diseases such as retinitis, esophagitis and colitis are mostly seen in immunocompromised patients like those with HIV infection or acquirea immunodeficiency syndrome

- Numerous drugs are available to treat herpesvirus infections

Thymidine analogues

- Example are Idoxuridine, Trifluridine
- Trifluridine is also referred as a fluorinated nucleoside

Mechanism of action

- Competes with thymidine leading to their incorporation into DNA
- This subsequently leads to the formation of faulty DNA that breaks easily

Problems associated with thymidine analogues:

- Low virus selectivity
- Rapid development of viral resistance

Indications of thymidine analogues

- Idoxuridine and Trifluridine are used in the treatment of the following conditions;

  - Herpes simplex keratitis (superficial)

NB: Trifluridine: higher efficacy

Side effects:

- Ocular irritation
- Lid oedema

Deoxyguanosine nucleoside analogues

Acyclovir

- This is a synthetic analogues of purine nucleoside called deoxyguanosine

Selective toxicity:

Unlike the thymidine analogues, deoxyguanosine analogues exhibit selective toxicity in the following ways

- Preferentially being taken up by virus infected cells
- Requiring only Herpes virus specific thymidine kinase for their action
- Acting by inhibiting viral DNA polymerase both reversibly and irreversibly

Mechanism of action for Acyclovir

- Acyclovir is a prodrug which is phosphorylated by virus encoded thymidine kinase to monophosphate metabolite and then later phosphorylated to active triphosphate metabolite by host kinases

- The active metabolite then competes with endogenous nucleoside triphosphates leading to competitive inhibition of DNA polymerase

- This eventually lead to prevention of viral DNA synthesis

- Acyclovir is also incorporated in the nascent DNA where it causes DNA chain termination while other nucleoside analogues like ganciclovir and penciclovir only inhibits viral DNA polymerase but do not cause DNA chain termination

Summary of the Mechanism of action for Acyclovir

Enzymes responsible drug phosphorylation;

1. Herpes virus specific thymidine kinase
2. Cellular kinases (host kinases)

Acyclovir

1

Acyclovir monophosphate

2

Acyclovir triphosphate

Competitively Inhibits herpes virus DNA polymerase

- Gets incorporated jnto viral DNA
- Early termination of lengthening of viral DNA
- DNA polymerase inhibited irreversibly

Pharmacokinetics of Deoxyguanosine analogues

- Acyclovir has oral bioavailability of 10-30%
- This decreases with increasing dose
- Valacyclovir has oral bioavailability of 55-70%

Acyclovir (Further properties)

- has low percutaneous absorption
- Widely distributed
- Eliminated un-metabolized by glomerular filtration and tubular secretion with elimination t½ of 2.5 hrs

- Acyclovir has low bioavailability and hence valacyclovir has been developed which is rapidly converted to acyclovir by intestinal and hepatic enzymes and has absorption than acyclovir

Indications of Acyclovir

Acyclovir are used in the following conditions;

Herpes simplex infections

- Genital (Primary, Recurrent)
- Mucocutaneous
- Herpetic encephalitis
- Keratitis

Varicella Zoster infections

- Herpes zoster
- Chicken pox

- Acyclovir and valacylovir can be used for prophylaxis of CMV infections such as in bone marrow or organ transplant recipients and in persons with HIV infection

Effects of drugs in treatment of HSV

When acyclovir, famciclovir or valacyclovir is given orally for treatment of herpes genitalis, the following are achieved;

- Prevention of replication of HSV
- Reduction of pain and other symptoms associated with acute infection
- Shortens the time to healing of lesions
- Reduces the amount of viral shedding
- It should be noted that HSV infections recurrences are common because they do not remove the viral copies

Effects of drugs in treatment of Shingles

When acyclovir, famciclovir or valacyclovir is given orally for treatment of shingles, the following is achieved;

- Shortening the duration of acute illness
- Shortening the duration of pain
- Shortening the duration of post-herpetic pain(Neuralgia)
- Famciclovir and valacyclovir are more effective than acyclovir in the treatment of shingles

Adverse effects of Acyclovir

- Dose dependent decrease in GFR
- Reversible neurological manifestation (lethargy, disorientation, hallucinations, convulsions, coma) at higher doses
- Topical: Stinging and burning sensation after each application
- Oral: Headache, nausea, malaise
- Intravenous: Rashes, sweating, fall in BP

Gancyclovir

- Analogue of acyclovir

Active against:

- H. simplex, H. zoster, EBV
- Cytomegalovirus (CMV, most sensitive)

- This needs a Higher concentration due to slow elimination which is greater than >24 hours

Ganciclovir cont'd

Mechanism of action

- Similar to acyclovir

Mechanism of resistance

- Mutation of viral phosphokinase and or viral DNA polymerase

Pharmacokinetics:

- Oral bioavailability is <10%
- Excreted in urine while the plasma half life 2-4 hrs
- Valganciclovir on the other hand has higher oral bioavailability and is about 100 times more active against CMV than acyclovir

Indications of Gancyclovir

- Prophylaxis and treatment of severe CMV infections in immunocompromised
- CMV retinitis in AIDS patients

Valganciclovir

- Valganciclovir is a prodrug that has better oral of ganciclovir with a better oral bioavailability and its converted to ganciclovir in the body

- It is approved for prevention of and treatment of CMV infections including those occurring in the renal and heart transplant patient

Adverse drug effects of ganciclovir

- Ganciclovir produces a higher incidence of adverse effects than acyclovir and famciclovir the common ones being;
- Leukopaenia and thrombocytopaenia
- Retinal detachment
- Liver and renal dysfunction
- Rash
- Fever
- Gastro intestinal disturbance

Famciclovir

- Famciclovir is a Guanine nucleoside
- It is a Prodrug of penciclovir
- Good oral bioavailability and actually has the highest bioavailability among the nucleoside analogues
- Prolonged intracellular t½ of the active triphosphate metabolite
- Penciclovir available in some countries for intravenous use

Mechanism of action of Famciclovir

- Requires viral thymidine kinase for activation
- Inhibits viral DNA polymerase

Indications

- Used as acyclovir sensitive H. simplex and H. zoster
- Reduces duration of illness
- Hepatitis B virus and mostly used in chronic hepatitis B

Side effects of Famciclovir

- Headache
- Mental confusion
- Nausea and loose motions
- Itching and rashes

Cidofovir

- Monophosphate nucleotide analogue of cytidine

Mechanism of Action:

- Does not require viral phosphokinase
- Converted into diphosphate by cellular kinase which remains intracellularly for long time
- Acts as alternative substrate for viral DNA polymerase and inhibits viral DNA polymerase

Indications:

- CMV retinitis (who have failed on ganciclovir therapy)
- Acyclovir resistant mucocutaneous HSV infection in immunocompromised
- Anogenital warts: topical

Pharmacokinetics of cidofovir

- Given as intravenous infusion along with pre and post dose oral probenecid which helps to achieve the following;
- Decrease tubular secretion of cidofovir
- Helps cidofovir to be readily available to enter cells for longer duration
- Reduces nephrotoxicity

Side effects

- Dose related kidney damage Neutropenia, uveitis, hypersensitivity reaction
- Cidofovir causes nephrotoxicity which is dose related
- Neutropenia
- Metabolic acidosis

- Cidofovir is therefore, reserved for infections that are resistant to ganciclovir and other drugs

- Cidofovir is further contraindicated in patients taking other nephrotoxic drugs such as aminoglycosides or amphotericin B

Other drugs used for herpes virus infections

FOSCANET

- This is a pyrophosphate derivative that blocks the pyrophosphate binding sites on the viral DNA polymerase thus preventing the binding of nucleotide precursors to DNA
- Foscanet does not need require activation by viral or host cell kinases
- It is active against CMV, VZV and HSV and is used to treat CMV retinitis in patients with AIDS and to treat acyclovir resistant HSV infections and shingles
- Foscanet and ganciclovir are also given in combination to treat infections that are resistant to either drug alone

Adverse effects of Foscanet

The following are the common adverse reactions of caused by foscanet;

- Renal impairment and acute renal failure
- Hematologic deficiencies
- Cardiac arrhythmias and cardiac failure
- Pancreatitis
- Due to hypocalcaemia:

  - Numbness and tingling sensation
  - Muscle cramps
  - Irritability
  - Seizures

- Renal toxicity can be minimized by administering intravenous fluids which help in inducing diuresis before and during foscanet

Conclusion

- Selective toxicity for antiviral agents is achieved by targeting viral enzymes and virus specific steps
- Anti-viral agents are broadly classified into four groups depending upon their activity on viruses
- Idoxuridine and trifluridine are used only topically
- Variation in the sensitivity of anti-herpes agents to herpes family viruses is well recognized

END